P07

An Eulerian Path Approach to DNA Fragment Assembly

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Domain DOM_BIO

Abstract

Fragment assembly is reformulated from finding a Hamiltonian path in an overlap graph to finding an Eulerian path in a de Bruijn graph. This transition reduces computational complexity and eliminates repeat assembly traps.

Abstract quoted from P. A. Pevzner et al., PNAS (2001).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?